

Gagan Jain

AI Engineer — GenAI · LLM · Agentic AI

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Agentic AI · RAG Pipelines · LLM Fine-tuning (LoRA/QLoRA) · Multi-Agent Systems · MCP Tooling · Prompt & Context Engineering · LangChain · LlamaIndex · Vector Databases · OpenAI / Anthropic APIs · LiteLLM · Docker · CI/CD

SUMMARY

AI Engineer building production-grade GenAI systems: agentic AI orchestration, RAG pipelines, LLM fine-tuning (QLoRA/LoRA), MCP tooling, and local-first LLM deployment. Author of SheshAOS, a governance-first, event-sourced AI operating system in Rust (9 workspace crates + CLI, 877+ tests, full CI/CD) with provider-agnostic LLM streaming, policy-enforced agent actions, and terminal/SSH integration. Strong Python + Rust foundation with end-to-end deployment: Docker, GitHub Actions, observability, and AI governance.

SKILLS

AI / LLM: LLM Integration, RAG Pipelines, Agentic AI Frameworks, Prompt Engineering, Context Engineering, Multi-Agent Systems, Tool-Using Agents, Fine-tuning (QLoRA/LoRA), MCP / Tool Protocols, LangChain / LlamaIndex, Vector Databases, AI Governance

Languages: Python, TypeScript, JavaScript, Rust, C++, C, PHP, HTML/CSS

Databases & DevOps: MySQL, MongoDB, Redis, SQLite, Supabase, Docker, GitHub Actions, Nginx, Unicorn, Prometheus, Linux

LLM Ops: Evaluation & Observability, Local LLM Deployment, OpenAI / Anthropic APIs, Embeddings, Structured Outputs / Function Calling, LiteLLM / Model Routing, Streaming Responses, NLP, Ollama

Backend & APIs: Flask, FastAPI, Node.js, Express, Laravel, Axum (Rust), REST APIs, WebSockets

Frontend & Vision: React, React Native, Next.js, Tailwind CSS, Bootstrap, Monaco Editor, YOLO, OpenCV, Computer Vision

PROJECTS

SheshAOS

AI / AGENTIC OS

Rust · Tokio · Event Sourcing · LiteLLM · OpenAI · Anthropic · SSH

- Built a governance-first, event-sourced AI OS in Rust — 9 workspace crates + CLI, 877+ passing tests, 0 clippy warnings, full GitHub Actions CI/CD
- Engineered a policy-enforced agent kernel where LLMs propose actions and the kernel validates/records every state change in an append-only audit trail
- Implemented a provider-agnostic LLM layer with OpenAI-compatible and Anthropic streaming, LiteLLM routing, and fully offline local-first inference
- Delivered native terminal emulation (PTY + VT100), SSH multiplexing, a secrets vault, and 4 interfaces: CLI, TUI, GUI, RPC

[Source Code](#) · [Documentation](#)

shesh-ecosystem

AGENTIC BODY

Python · Rust · MCP · Agentic AI · Ollama · Governance

- Built a federated, local-first AI body for CachyOS/Hyprland — Brain (governance kernel), Mind (models/planning/memory), Soma (sensors/actuators)
- Designed the orchestrator manifest: 22 components, 3 channels (stable/canary/devel), SHA256-audited locks, 30-test ecosystem gate
- Enabled local-first inference — Ollama phi4-mini / qwen2.5-coder:3b / moondream2 / nomic-embed-text in 6GB VRAM, fully offline
- Enforced governance via shesh-audit GuardedMCP policy (allow/confirm/deny) + hash-chained audit + Kernel bridge; swarm through GitHub Issues atomic locks

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Vyakrti

COMPILER

Rust · React · TypeScript · Monaco · Axum

- Built a complete compiler pipeline from scratch in Rust: lexer → parser → type checker → bytecode compiler
- Developed a browser-based IDE (React + Monaco Editor) with syntax highlighting, autocomplete, and diagnostics
- Implemented a Rust (axum) backend with compile, REPL, LSP, and file management via REST + WebSocket
- Wrote 123 tests covering the full pipeline, including a self-hosting corpus

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rag-service

RAG / VECTOR

Python · FastAPI · ChromaDB · Hybrid Search · LLM-as-Judge

- Built a FastAPI RAG service with hybrid retrieval (dense embeddings + BM25, merged via Reciprocal Rank Fusion) over ChromaDB
- Implemented chunking strategies (recursive, semantic) and embedding pipelines; /ask endpoint streams grounded answers with citations
- Shipped an LLM-as-judge evaluation harness (faithfulness, answer relevance, correctness) with a golden-set CI gate
- Containerized with Docker Compose; OpenAI-compatible LLM interface with local-first fallback

[Source Code](#) · [Documentation](#)

llm-eval-harness

LLM EVAL

Python · LLM-as-Judge · Golden Sets · RAGAS · Pytest

- Built a golden-set-driven evaluation harness with faithfulness, answer-relevance, and correctness metrics (LLM-as-judge + lexical fallbacks)
- Added offline heuristic scorers so evals run in CI without API keys; JSON + Markdown reports
- Designed a golden-set YAML format (question, golden answer, context) and a CLI (python -m eval_harness)
- Containerized and CI-ready; used to gate the RAG service pipeline

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AIM

PRODUCTION

Python · Flask · MySQL · Bootstrap · Chart.js · Docker

- Built a production-grade Flask + MySQL platform: Argon2id auth, CSRF protection, brute-force lockout, breached-password scanning, JWT sessions, strict CSP/HSTS
- Shipped Prometheus metrics, structured JSON logging, Chart.js analytics, FullCalendar scheduling, Docker Compose, and CI/CD — 101 automated pytest tests

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FWRS

PRODUCTION

Python · PuLP · Folium · Flask · MySQL

- Built a 3-stage lexicographic LP solver (fairness → priority → cost) that allocates surplus food to NGOs while minimizing waste
- Added expiry-aware routing that penalizes allocations where travel time exceeds food shelf life
- Integrated Folium/Leaflet interactive maps with animated routes, heatmaps, and priority-colored markers
- Shipped a Flask web dashboard plus a desktop Tkinter GUI with charts and CSV export

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shesh-omniroute

AI GATEWAY

TypeScript · OmniRoute · AI Gateway · Free Models · MCP · A2A

- Wrapped OmniRoute — a free, MIT-licensed AI gateway (291 providers, 90+ free, 500+ models) — as a Shesh component, optional alongside a local Ollama primary
- Documented 1.53B free tokens/month; RTK+Caveman compression cuts 15–95% of tokens (~89% average)
- Exposed one OpenAI-compatible endpoint (http://localhost:20128/v1) with auto-fallback across tiers: Sub → API → Cheap → Free
- Surfaced 19 routing strategies, 105 MCP tools, A2A v0.3, Desktop/PWA, 43 i18n locales, MIT self-hosted

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EXPERIENCE

Industry 5.0 Industrial Automation Trainee

Aug 2025 – Nov 2025

[CodenPlay Robotics](#) · [VIT Vellore](#) / [Online](#)

- Built and debugged PLC automation logic (ladder logic, structured text) in CODESYS
- Simulated industrial process workflows in Factory I/O
- Implemented flow-based control/integration in Node-RED
- Covered HMI design, industrial communication, process monitoring, and the debugging → testing → deployment workflow

EDUCATION

Vellore Institute of Technology, Vellore

Graduated 2025

[B.Tech in Computer Science and Engineering](#)

CGPA: 7.7 / 10.0 · Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Web Technologies, Artificial Intelligence, Machine Learning